

**U.S. Department of Homeland Security
U.S. Customs and Border Protection**



Statement of Work

For

**Office of Information and Technology (OIT)
Enterprise Infrastructure and Operations Directorate (EIOD)
Enterprise Wireless Communications Branch (EWCB)**

COTHEN TECHNOLOGY REFRESH

PR 2015-3254

January 26, 2026

**REQUEST FOR INFORMATION
ATTACHMENT 1 SOW**

Office of Information and Technology (OIT)
Enterprise Infrastructure and Operations Directorate (EIOD)
Enterprise Wireless Communications Branch (EWCB)
COTHEN TECHNOLOGY REFRESH
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1. BACKGROUND

U.S. Customs and Border Protection (CBP) developed and deployed the High Frequency (HF) Cellular Over-the-Horizon Enforcement Network (COTHEN) radio network in order to meet CBP's tactical long-range radio communications requirements. The program supports the Department of Homeland Security (DHS) as well as other federal agencies.

2. SCOPE

The purpose of this procurement is to procure a new Wide Band High Frequency Receiver-Transmitter to replace the Collins Aerospace URGIII High Frequency Communication System that is approaching end of lifecycle support by the vendor. The COTHEN HF Communication Network infrastructure contains 17 Fixed sites with one (1) radio system per site. The radio system requires interoperability with the current COTHEN High Frequency Cellular Network system that operates with Collins Aerospace software.

2.1 Brand Name or Equal Requirement

The radio system specified in this Statement of Work is the **Rockwell Collins RT-2200A Wide Band High Frequency Receiver-Transmitter Radio System, or equal**.

Offerors may propose an "equal" product, provided it meets or exceeds all of the salient physical, functional, and performance characteristics specified below and elsewhere in this SOW.

Offerors proposing an "equal" product must:

- Clearly identify the manufacturer, model, and specifications of the proposed system.
- Provide documentation and evidence demonstrating that the proposed system meets or exceeds all salient characteristics and is fully compatible with the COTHEN HF Cellular Network.
- The Government reserves the right to request additional information or samples for evaluation purposes.

The Government will evaluate "equal" products based on the information furnished by the offeror. If the offeror fails to provide sufficient information to demonstrate equivalence, the Government will determine that the offeror's product does not meet the requirements.

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This requirement is in accordance with Federal Acquisition Regulation (FAR) 52.211-6, Brand Name or Equal.

3. APPLICABLE DOCUMENTS

The Contractor shall adhere to the following applicable Federal standard /DHS/CBP/Office of Information and Technology (OIT) policies, standards, directives, processes, and procedures in performing work under this contract. These include, and are not limited to the following:

- The Federal Acquisition Regulation (FAR): <http://www.acqnet.gov/FAR/>
- Section 508 of the Rehabilitation Act: <http://www.section508.gov/>
- Homeland Security Acquisition Regulation (HSAR):
<http://www.dhs.gov/xlibrary/assets/opnbiz/hsar.pdf>
- Code of Federal Regulations (CFR): <http://www.gpoaccess.gov/cfr/index.html>
- Customs Directive No. 51715-006, Separation Procedures for Contractor Employees (CF-242)
- CIS HB 1400-05C, "CBP Information Systems Security Policies and Procedures Handbook"
- Management Directive (MD) 4300.1, "Information Technology Systems Security Program"
- "DHS Sensitive System Policy," Directive 4300A:
<http://www.uscg.mil/hq/cg9/NAIS/RFP/SectionJ/dhs-4300A-policy.pdf>
- NIST SP 800-47, "Security Guide for Interconnecting Information Technology Systems":
<http://csrc.nist.gov/publications/nistpubs/800-47/sp800-47.pdf>
- DHS 4300A, "Sensitive Systems Directive":
<http://www.uscg.mil/acquisition/nais/RFP/SectionJ/dhs-4300A-handbook.pdf>

4. SPECIFIC TASKS/TECHNICAL REQUIREMENTS

The Contractor shall complete the following tasks as required to comply with this SOW.

4.1 Salient Characteristics:

The replacement radio systems must, at a minimum, provide:

- Frequency range: 1.5 to 29.9999 MHz transmit-and-receive
- Output power: 1 kW (+/-1dB) PEP and average
- Duty cycle: 100%
- Modes: SSB, ISB, CW, AM, AME Tx, WBHF
- Data rates: Waveforms supporting from 75 bps up to 240 kbps in up to 16 x 3 kHz channels (48 kHz max. BW)
- Built-in 4G ALE
- Remote control via web browser or external remote-control software

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- Compliance with MIL-STD-188-110D and MIL-STD-188-141D
- Interoperability with the existing COTHEN High Frequency Cellular Network, including compatibility with Collins Aerospace software and protocols currently in use
- All other specifications and requirements listed in this SOW

4.2 Testing

The Contractor shall test and functionally certify all equipment prior to delivery to ensure compliance and specifications.

5. DELIVERABLES AND DELIVERY SCHEDULE

The Contractor shall deliver equipment as outlined below. The equipment shall be shipped to the address provided in section 5.1.

The following details deliverables that are to be provided under this Task Order.

Receiver-Transmitter Radio System:

- Delivers data rates over 20 times faster than legacy HF
- Waveforms support data rates 75 bps up 240 kbps in up to 16 x 3 kHz channels (48 kHz max. BW)
- Built-in 4G ALE for reliability and ease of use
- Supports flexible architecture, enabling networking capabilities in commercial and military installations
- Frequency: 1.5 to 29.9999 MHz transmit-and-receive frequency ranges in regular operation
- Simplex and half-duplex operation
- An electronically tuned RF bandpass filter intended for Cosite applications
- SSB, two-channel ISB, CW, AM, AME Tx, WBHF modes
- Narrowband and wideband HF compliant per MIL-STD-188-110D and MIL-STD-188-141D Provides analog voice
- Chat – sending and receiving text messages over HF
- File transfer – sending and receiving files up to 20 times faster than legacy HF
- Remote control via web browser, or external remote-control software
- Software updates/datafill: via USB on front panel or remote via Ethernet

RF Output:

- Output power: Per MIL-STD 188-141D 5.3.7
 - 1 kW (+/-1dB)
 - PEP and average up to 1.3:1 VSWR
 - Max VSWR: 3:1

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- Duty cycle: 100%
- Output impedance: 50 Ohms at 1.5:1 VSWR
- Emissions compliance: Per MIL-STD-188-141D
 - 5.2.7.1-, 5.2.7,
 - 5.3.1.1, 5.3.1.2, 5.3.1.3, 5.3.2.3
- Antenna connector N-type female

Control:

- Ethernet: Remote HTML or remote-control software
- Serial RS-232 and RS-422: Remote-software control
- Interlock: Antenna interlock to prevent keying of PA when not connected
- Data interfaces, Ethernet, RS-232, EIA-530 (synchronous serial, asynchronous serial)

Input Power:

- 110 - 264 VAC single phase, 47-63 Hz

Temperature:

- Operating: -4° F to 122° F (-20° C to 50° C)
- Storage: -40° F to 158° F (-40° C to 70° C)

Humidity:

- Humidity: 95% non-condensing

5.1 Delivery

Attn: Terry Horton
Department of Homeland Security
CBP, OIT, NLECC, EWCD, COTHEN
Address: 1900 Lakemont Ave, Orlando, FL 32803
Phone Number: 407-975-1974
Terry.t.horton@cbp.dhs.gov

The Government Purchase Order number must be reflected on the Shipping label and clearly delineated on the packing slip(s).

5.2 Product Description and specifications-Brand Name or Equal

Rockwell Collins RT-2200A Wide Band High Frequency Receiver Transmitter Radio

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5.3 Bill of Materials

MFG P/N	Manufacturer	Description	QTY
Brand Name or Equal RT- 2200A	Brand Name or Equal to Collins Aerospace (formerly Rockwell Collins)	Brand Name or Equal to Rockwell Collins RT- 2200A Wide Band HF Radio	5
		Cables, connectors, and mounting hardware	5
		Proprietary Software	5
		Shipping & Handling	1
		Warranty/Support (1 year)	1

6. GOVERNMENT FURNISHED EQUIPMENT AND INFORMATION

The Government does not anticipate providing any Government Furnished Equipment (GFE) or Government Furnished Information (GFI) under this award.

7. PLACE OF PERFORMANCE

The delivery of goods will be provided to the following CBP location:

CBP/NLECC
1900 Lakemont Ave.
Orlando, FL 32803
ATTN: Terry Horton

8. PERIOD OF PERFORMANCE

The period of performance is Date of Award (DOA) for Six Months.

9. SECURITY

There are no additional security classification level requirements associated with this procurement action, other than those listed elsewhere in this SOW. Information is deemed unclassified.

10. SPECIAL CONSIDERATIONS

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10.1 Quality Assurance/Acceptance

The Contracting Officer's Representative (COR) or designees review, for completeness, final work products and deliverables that the Contractor submits, and may return them to the Contractor for correction. Absence of any comments by the COR does not relieve the Contractor of the responsibility for complying with the requirements of the Task Order.

10.2 Government Acceptance Period

The COR will review deliverables prior to acceptance. If the delivery is not acceptable, the COR will document and provide reason(s) to the Contractor in writing. The COR will have ten (10) business days to review the deliverables and make comments. The Contractor shall have five (5) business days to make corrections and redeliver. Should the Government fail to complete the review within the review period the deliverable will be considered "Accepted".

10.3 Travel

Travel is not required to support the tasks outlined in this SOW.

10.4 Task Order Administration

Notwithstanding the contractor's responsibility for management during performance the administration of this task order will require some coordination between the CBP and the contractor. The COR will assure orderly performance of the tasks and provide technical direction. The COR is:

Tamika Shung
Department of Homeland Security
CBP, Office of Information and Technology, EWCB
Address: 1900 Lakemont Ave, Orlando, FL, 32803
Phone: 1 (407)-975-1725
Tamika.Shung@cbp.dhs.gov

The types of actions within the purview of the COR's authority are to assure that the contractor performs the technical requirements of the contract; to perform or cause to be performed inspections necessary in connection with performance of the contract; to maintain both written and oral communications with the contractor concerning the aspects of the contract within his/her purview; to issue interpretations of technical requirements; to monitor the contractor's performance under the task order and notify the contractor and CO of any deficiencies observed; and to coordinate Government-Furnished Property or Data availability and provide for site entry of contractor personnel if required.

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The COR will provide no supervision to contractor personnel. The CO is the only one empowered to make any commitments or changes which affect the task order price or other items and conditions. Any such proposed changes must be brought to the immediate attention of the CO for action. The acceptance of any changes by the contractor without specific approval and written consent of the CO shall be at the contractor's risk.

10.5 Invoice Instructions

Payment requests shall be submitted electronically through the U. S. Department of the Treasury's Invoice Processing Platform System (IPP).

The IPP website address is: <https://www.ipp.gov>.

If the Contractor is unable to comply with the requirement to use IPP for submitting invoices for payment, the Contractor must submit a waiver request in writing to the Contracting Officer.

Please ensure the following are included in all invoices:

- Purchase Order number
- Description of services provided for a specified time period.
- Unit price and total amount of each item.
- Discount terms
- Company name, telephone number, taxpayer's identification number, and complete mailing address to which payment will be mailed.

A courtesy e-mail notification upon invoice submission in IPP shall be sent by the Contractor to the COR.

Only the CO has the authority to represent the Government in cases where the task order requires a change in the terms and conditions, delivery schedule, scope of work and/or price of the products and/or services under this task order.

10.6 Points of Contact and Task Order Administration

All contract questions and concerns shall be directed to the point of contracts as designated below. Only the Contracting Officer (CO) has the authority to represent the Government in cases where the task order requires a change in the terms and conditions, delivery schedule, scope of work and/or price of the products and/or services under this task order.

COR:

Tamika Shung

Department of Homeland Security

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Address: 1900 Lakemont Ave, Orlando, FL, 32803

Phone: 1 (571) 468-0887

Michelle.K.Smith@cbp.dhs.gov

Technical POC:

Name: Terry Horton

Department of Homeland Security

CBP, Office of Information and Technology, EWCB

Address: 1900 Lakemont Ave, Orlando, FL, 32803

Phone: 1 (407)-975-1974

Terry.t.horton@cbp.dhs.gov

Local Property Officer:

Name: Mark Gonska

Department of Homeland Security

CBPOIT, EIOE, EWCB, COTHEN

Address: 1900 Lakemont Ave, Orlando, FL 32803

Phone: 1 (407) 975-1719

Fax: 1 (407) 975-2036

Email: Donna.Ward@cbp.dhs.gov

ADDENDUM A - ACRONYMS LIST

ACRONYMS	DESCRIPTION
AES	Advanced Encryption Standard
BOM	Bill of Material
CBP	Customs and Border Protection
CFR	Code of Federal Regulations
CO	Contract Officer
COB	Close of Business
COR	Contracting Officer's Representative
COTHEN	Cellular-over-The Horizon-Enforcement Network
DHS	Department of Homeland Security
DVP	Digital Voice Privacy
EWCB	Enterprise Wireless Communications Branch
FAR	Federal Acquisition Regulation
FIPS	Federal Information Processing Standards
FY	Fiscal Year
GFE	Government Furnished Equipment
GFI	Government Furnished Information
OIT	Office of Information and Technology
PMP	Program Management Plan
PoP	Period of Performance
PR	Purchase Requisition

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QA	Quality Assurance
RC	Rockwell Collins
SAS	Sierra Automated Systems
SB	Service Bulletin
SOW	Statement of Work
U.S.	United States

ADDENDUM B – TERMS AND CONDITIONS

CBP Enterprise Architecture Compliance

The Offeror shall ensure that the design conforms to the Department of Homeland Security (DHS) and Customs and Border Protection (CBP) Enterprise Architecture (EA), the DHS and CBP Technical Reference Models (TRM), and all DHS and CBP policies and guidelines (such as the CBP Information Technology Enterprise Principles and the [DHS Service Oriented Architecture - Technical Framework](#)), as promulgated by the DHS and CBP Chief Information Officers (CIO), Chief Technology Officers (CTO) and Chief Architects (CA).

The Offeror shall conform to the Federal Enterprise Architecture (FEA) model and the DHS and CBP versions of the FEA model, as described in their respective EAs. All models will be submitted using Business Process Modeling Notation (BPMN 1.1 or BPMN 2.0 when available) and the CBP Architectural Modeling Standards. Universal Modeling Language (UML2) may be used for infrastructure only. Data semantics shall be in conformance with the National Information Exchange Model (NIEM). Development solutions will also ensure compliance with the current version of the DHS and CBP target architectures.

Where possible, the Offeror shall use DHS/CBP approved products, standards, services, and profiles, as reflected by the hardware, software, application, and infrastructure components of the DHS/CBP TRM/standards profile. If new hardware, software, or infrastructure components are required to develop, test, or implement the program, these products will be coordinated through the DHS and CBP formal Technology Insertion (TI) process (to include a trade study with no

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less than four alternatives, one of which reflecting the status quo and another reflecting multi-agency collaboration). The DHS/CBP TRM/standards profile will be updated as TIs are resolved.

All developed solutions shall be compliant with the Homeland Security (HLS) EA.

All IT hardware and software shall be compliant with the HLS EA.

Compliance with the HLS EA shall be derived from and aligned through the CBP EA.

Description information for all data assets, information exchanges and data standards, whether adopted or developed, shall be submitted to the Enterprise Data Management Office (EDMO) for review, approval, and insertion into the DHS Data Reference Model and Enterprise Architecture Information Repository.

Development of data assets, information exchanges, and data standards will comply with the DHS Data Management Policy MD 103-01. All data-related artifacts will be developed and validated according to DHS Data Management Architectural Guidelines.

Applicability of Internet Protocol version 6 (IPv6) to DHS-related components (networks, infrastructure, and applications) specific to individual acquisitions shall be in accordance with the DHS EA (per OMB Memorandum M-05-22, August 2, 2005), regardless of whether the acquisition is for modification, upgrade, or replacement. All EA related component acquisitions shall be IPv6 compliant, as defined in the USGv6 Profile (NIST Special Publication 500-267) and the corresponding declarations of conformance, defined in the USGv6 Test Program.

Section 508 Requirements

Section 508 of the Rehabilitation Act (classified to [29 U.S.C. § 794d](#)) requires that when Federal agencies develop, procure, maintain, or use information and communications technology (ICT), it shall be accessible to people with disabilities. Federal employees and members of the public with disabilities must be afforded access to and use of information and data comparable to that of Federal employees and members of the public without disabilities.

All products, platforms and services delivered as part of this work statement that, by definition, are deemed ICT shall conform to the revised regulatory implementation of Section 508 Standards, which are located at 36 C.F.R. § 1194.1 & Appendixes A, C & D, and available at <https://www.ecfr.gov/cgi-bin/text-idx?SID=e1c6735e25593339a9db63534259d8ec&mc=true&node=pt36.3.1194&rgn=div5>. In the revised regulation, ICT replaced the term electronic and information technology (EIT) used in the original 508 standards. ICT includes IT and other equipment.

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Exceptions for this work statement have been determined by DHS and only the exceptions described herein may be applied. Any request for additional exceptions shall be sent to the Contracting Officer and a determination will be made according to DHS Directive 139-05, Office of Accessible Systems and Technology, dated November 12, 2018 and DHS Instruction 139-05-001, Managing the Accessible Systems and Technology Program, dated November 20, 2018, or any successor publication.

Section 508 Applicable Exceptions

Section 508 applicability to Information and Communications Technology (ICT): Receiver Transmitter Radio System

Applicable Exception: National Security Authorization #: CBP-20150217-001

Exceptions for this work statement have been determined by DHS and only the exceptions described herein may be applied. Any request for additional exceptions shall be sent to the Contracting Officer and a determination will be made according to DHS Directive 139-05, Office of Accessible Systems and Technology, dated November 12, 2018 and DHS Instruction 139-05-001, Managing the Accessible Systems and Technology Program, dated November 20, 2018, or any successor document.

Compliance with DHS Security Policy

All hardware, software, and services provided under this task order must be compliant with DHS 4300A DHS Sensitive System Policy and the DHS 4300A Sensitive Directive Handbook.

Encryption Compliance:

If encryption is require, the following methods are acceptable for encrypting sensitive information:

1. FIPS 197 (Advanced Encryption Standard (AES)) 256 algorithm and cryptographic modules that have been validated under FIPS 140-2.
2. National Security Agency (NSA) Type 2 or Type 1 encryption.
3. Public Key Infrastructure (PKI) (see paragraph 5.5.2.1 of the Department of Homeland Security (DHS) IT Security Program Handbook (DHS Management Directive (MD) 4300A) for Sensitive Systems).